

Book Qs.

Q1.

ü $S \rightarrow aSbb \mid abb$

$S \rightarrow ASBB \mid ABB$

$S \rightarrow AR_1$

$R_1 \rightarrow SBB \quad S \rightarrow AR_2$

$R_1 \rightarrow SR_2$

$R_2 \rightarrow BB$

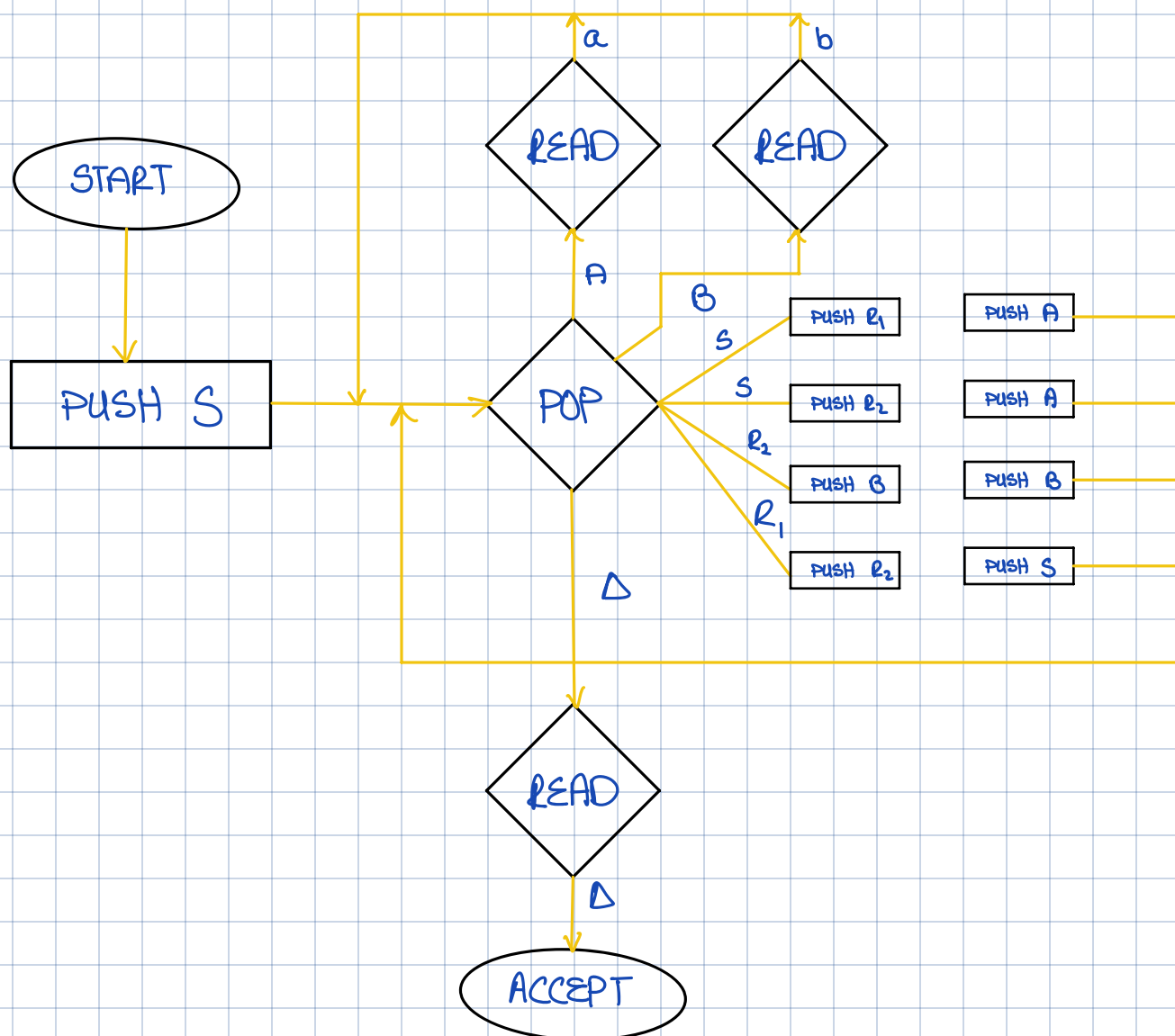
$S \rightarrow AR_1 \mid AR_2$

$R_1 \rightarrow SR_2$

$R_2 \rightarrow BB$

$A \rightarrow a$

$B \rightarrow b$



Q3.

$$S \rightarrow aS \mid aSbS \mid a$$

$$S \rightarrow AS \mid ASBS \mid a$$

$$S \rightarrow ASBS$$

$$S \rightarrow AR_1$$

$$R_1 \rightarrow SBS$$

$$R_1 \rightarrow SR_2$$

$$R_2 \rightarrow BS$$

Now,

$$S \rightarrow AS \mid AR_1 \mid a$$

$$R_1 \rightarrow SR_2$$

$$R_2 \rightarrow BS$$

Q2.

$$S \rightarrow XaaX$$

$$X \rightarrow aX \mid bX \mid \Lambda$$

$$X \rightarrow aX$$

$$\Rightarrow a\Lambda$$

$$\Rightarrow a$$

$$X \rightarrow bX$$

$$\Rightarrow b\Lambda$$

$$\Rightarrow b$$

$$S \rightarrow XaaX$$

$$\Rightarrow \Lambda aa \Lambda$$

$$\Rightarrow aaX$$

$$\Rightarrow Xaa\Lambda$$

$$\Rightarrow Xaa$$

$$\Rightarrow \Lambda aa\Lambda$$

$$\Rightarrow aa$$

$$S \rightarrow XaaX \mid aaX \mid Xaa \mid aa$$

$$X \rightarrow aX \mid bX \mid a \mid b$$

$$S \rightarrow XAAX \mid AAX \mid XAA \mid AA$$

$$X \rightarrow AX \mid BX \mid a \mid b$$

$$S \rightarrow XAAX$$

$$S \rightarrow XR_1$$

$$R_1 \rightarrow AAX$$

$$R_1 \rightarrow AR_2$$

$$R_2 \rightarrow AX$$

$$S \rightarrow AAX$$

$$S \rightarrow AR_2$$

$$S \rightarrow XAA$$

$$S \rightarrow XR_3$$

$$R_3 \rightarrow AA$$

$$S \rightarrow XR_1 \mid AR_2 \mid XR_3 \mid AA$$

$$R_1 \rightarrow AR_2$$

$$R_2 \rightarrow AX$$

$$R_3 \rightarrow AA$$

$$A \rightarrow a$$

$$B \rightarrow b$$

$$X \rightarrow AX \mid BX$$

Q6.

$$S \rightarrow Saa \mid aSa \mid aaS$$

$$S \rightarrow SAA \mid ASA \mid AAS$$

$$R_1 \rightarrow AA$$

$$R_2 \rightarrow SA$$

$$Q7. S \rightarrow (S)(S) \mid a$$

$$S \rightarrow LSR \mid LSR \mid a$$

$$R_1 \rightarrow SR$$

$$S \rightarrow LR_1 \mid LR_1$$

$$R_2 \rightarrow LR_1$$

Now,
 $S \rightarrow SR_1 \mid AR_2 \mid R_1S$
 $R_1 \rightarrow AA$
 $R_2 \rightarrow SA$
 $A \rightarrow a$

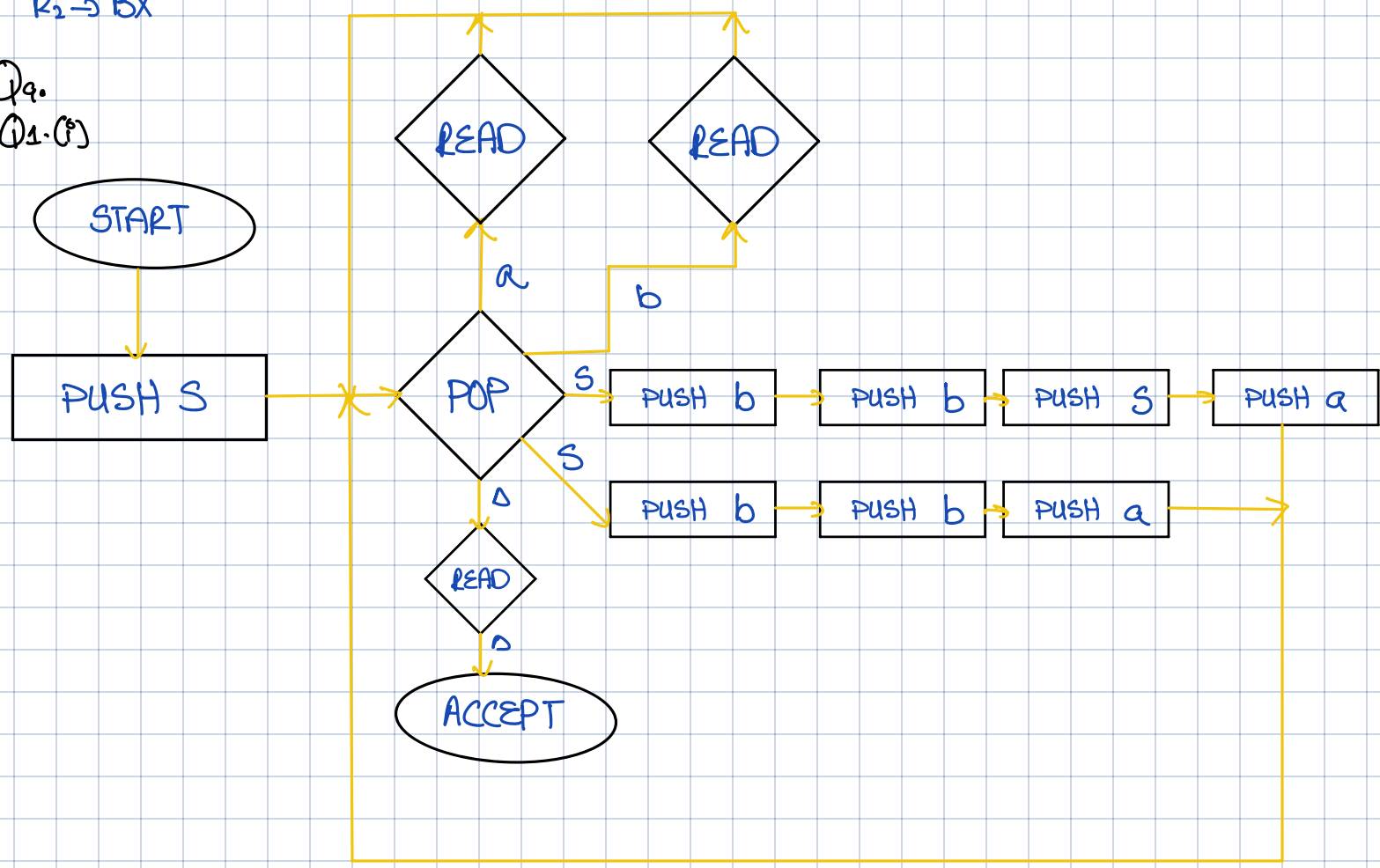
$S \rightarrow R_2R_2 \mid a$
 $R_1 \rightarrow SR$
 $R_2 \rightarrow LR_1$
 $L \rightarrow C$
 $R \rightarrow J$

Q8.
 (i) $S \rightarrow XaY \mid YbX$
 $X \rightarrow YY \mid aY \mid b$
 $Y \rightarrow b \mid bb$

$S \rightarrow XAY \mid YBX \rightarrow S \rightarrow XAY \quad S \rightarrow YBX$
 $X \rightarrow YY \mid AY \mid b \quad S \rightarrow XR_1 \quad S \rightarrow YR_2$
 $Y \rightarrow b \mid BB \quad R_1 \rightarrow AY \quad R_2 \rightarrow BX$

Now,
 $S \rightarrow XR_1 \mid YR_2$
 $X \rightarrow YY \mid AY \mid b$
 $Y \rightarrow b \mid BB$
 $R_1 \rightarrow AY$
 $R_2 \rightarrow BX$

Q9.
 Q1. (*)



✓

START

ACCEPT

REJECT

READ

POP

PUSH

Q15.

i) $S \rightarrow aSb \mid ab$

$S \rightarrow AR \mid AB$

$R \rightarrow SB$

$A \rightarrow a$

$B \rightarrow b$